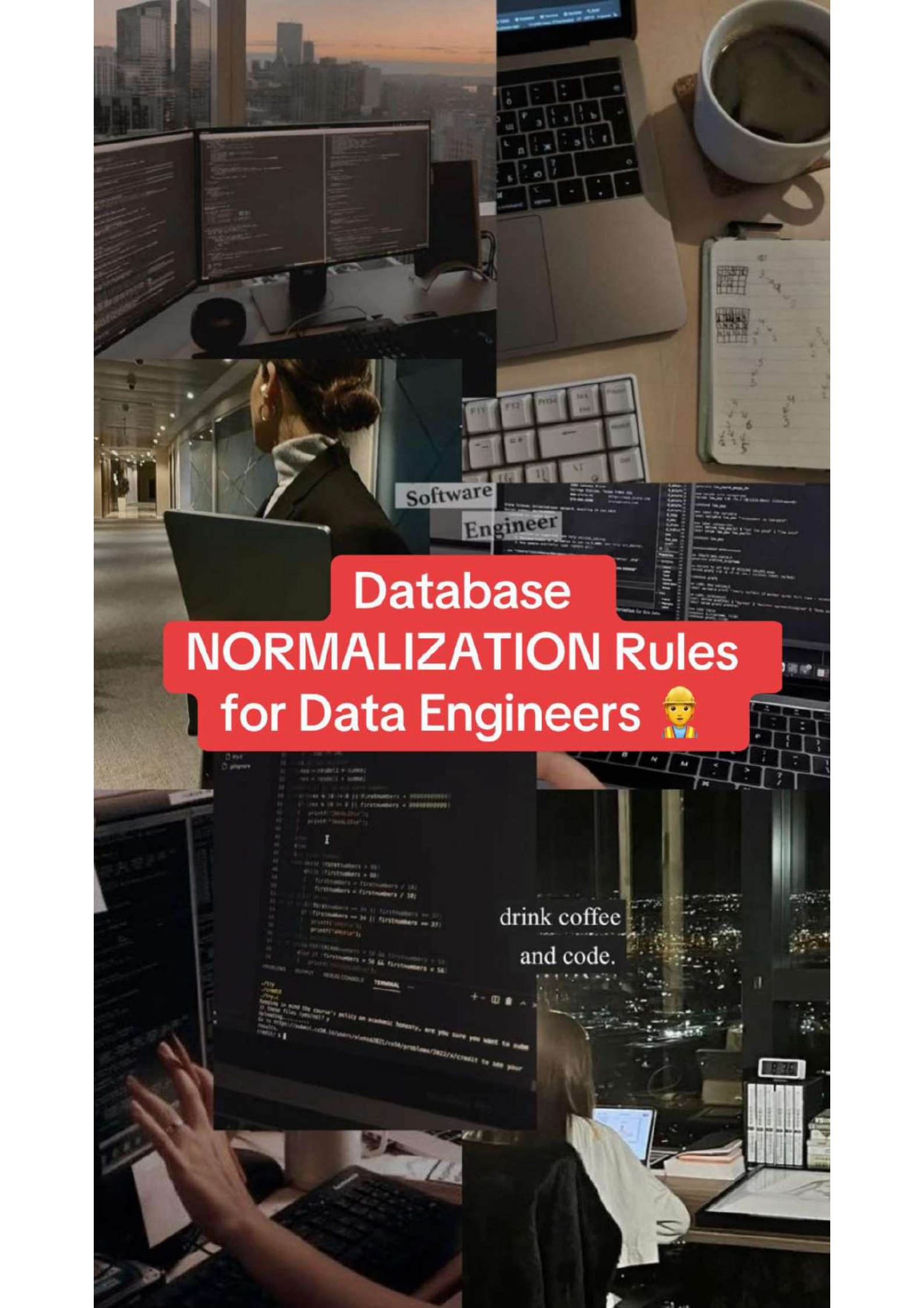




Database NORMALIZATION Rules for Data Engineers 🧑‍💻

drink coffee
and code.

This table doesn't work

OrderID	Customer	Parts_Purchased	Manufacturer	Manufacturer_Country
101	Mario Rossi	GPU, RAM	NVIDIA, Corsair	USA, USA
102	Luca Bianchi	CPU	Intel	USA



✗: Each table cell must contain a single (atomic) value, and each record must be unique.

1 1st Normal Form (1NF): Atomicity



OrderID (PK)	Part (PK)	Customer	Manufacturer	Manufacture- r_Country
101	GPU	Mario Rossi	NVIDIA	USA
101	RAM	Mario Rossi	Corsair	USA
102	CPU	Luca Bianchi	Intel	USA

**Problem: All non-key
columns must depend on
the entire Primary Key.**

2 2nd Normal Form (2NF): Full Dependency

OrderID (PK)	Customer
101	Mario Rossi
102	Luca Bianchi

OrderID (FK)	Part (PK)	Manufacturer	Manufacturer_ Country
101	GPU	NVIDIA	USA
101	RAM	Corsair	USA
102	CPU	Intel	USA

**Problem: Non-key
attributes must depend
on the key, the whole key,
and nothing but the key**

3 3rd Normal Form (3NF): No Transitive Dependency

OrderID (PK)	Customer
101	Mario Rossi
102	Luca Bianchi

Part (PK)	ManufacturerID (FK)
GPU	M1
RAM	M2
CPU	M3

ManufacturerID (PK)	Manufacturer_Name	Country
M1	NVIDIA	USA
M2	Corsair	USA
M3	Intel	USA

Correct form 